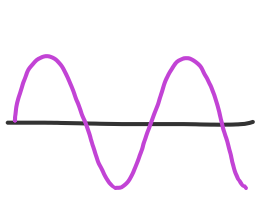
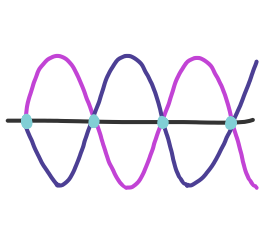
Matter Waves

Monday, November 10, 2025 3:53 PM

Travelling wave : $y(x,t) = A\sin(kx-\omega t)$



Standing wave : $y(x,t) = 2A\sin(kx)\cos(\omega t)$



Fourier series and wave packets :

any localized wave as superposition of many sinusoidal waves :

$$\boxed{\begin{aligned} y(x,t) &= \sum A_k \sin(kx-\omega_k t) \\ y(x,t) &= \int \underset{\uparrow}{A(k)}\, e^{i(kx-\omega t)}\,dk \end{aligned}}$$

A is a measurable quantity like displacement, pressure, electric field

express using complex notation: $y(x,t) = \mathrm{Re}\big[\underline{Ae^{i(kx-\omega t)}}\big]$ (Euler's formula: $e^{i\phi} = \cos\phi + i\sin\phi$)

$A(\cos(kx-\omega t) + i\sin(kx-\omega t))$

classical mechanics :

Wave has real and imaginary part : $\Psi = \underset{\substack{\uparrow \\ \text{physical motion}}}{\Psi_{real}} + \underset{\substack{\uparrow \\ \text{mathematical helper}}}{\Psi_{imag}}$ (classical!)

Overall magnitude : $|\Psi| = \sqrt{\Psi_{real}^2 + \Psi_{imag}^2} = A$

quantum mechanics :

Ψ encodes all mathematical information of a system – position, momentum, energy, angular momentum, etc.

Ψ is a probability amplitude, not a physical wave

$\underset{\substack{\uparrow \\ \text{real, positive}}}{|\Psi|^2} = \Psi^*\Psi$ is probability density of finding particle at r at time t

amplitude - phase form : $\boxed{\Psi(x,t) = R(x,t)\, e^{i\phi(x,t)}}$

ϕ determines momentum, interference, time evolution

If $|\Psi|^2$ is the probability density, then Ψ must satisfy the following :

① continuous, single-valued, finite

② continuous first-order derivative

③ normalizable

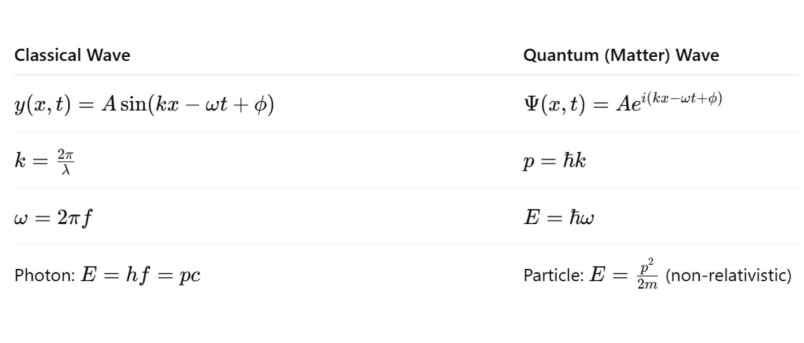
$$\boxed{\int_{-\infty}^{\infty} |\Psi(x)|^2\,dx = 0}$$ $\Psi(x)$ normalizable

same sinusoidal form can represent both light and matter waves

$E = \hbar\omega$

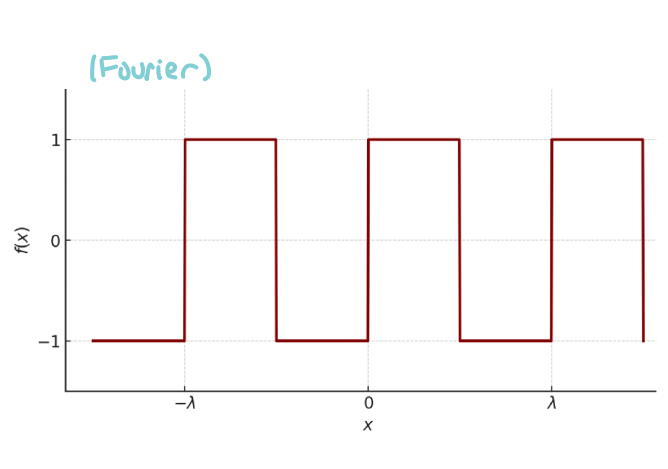
$p = \frac{h}{\lambda} = \hbar k$

$$\Psi(x,t) = Ae^{i(kx \mp \omega t + \phi)} = Ae^{\frac{i}{\hbar}(px \mp Et + \phi)}$$



Classical Wave	Quantum (Matter) Wave
$y(x,t) = A\sin(kx-\omega t+\phi)$	$\Psi(x,t) = Ae^{i(kx-\omega t+\phi)}$
$k = \frac{2\pi}{\lambda}$	$p = \hbar k$
$\omega = 2\pi f$	$E = \hbar\omega$
Photon: $E = hf = pc$	Particle: $E = \frac{p^2}{2m}$ (non-relativistic)

Ex. (Fourier)



$f(x) = 1, \quad 0 < x < \frac{\lambda}{2}$

$f(x) = -1, \quad \frac{\lambda}{2} < x < \lambda$

odd function $\Rightarrow$ $A_n = 0$, $B_n = ?$

$$\begin{aligned} B_n &= \frac{2}{\lambda}\left[\int_0^{\lambda/2} (1)\sin\frac{2\pi n x}{\lambda}\,dx + \int_{\lambda/2}^{\lambda} (-1)\sin\frac{2\pi n x}{\lambda}\,dx\right] \\ &= \frac{2}{\lambda}\left[-\frac{\lambda}{2\pi n}\cos\frac{2\pi n x}{\lambda}\right]_0^{\lambda/2} + \frac{2}{\lambda}\left[\frac{\lambda}{2\pi n}\cos\frac{2\pi n x}{\lambda}\right]_{\lambda/2}^{\lambda} \\ &= -\frac{1}{n\pi}(\cos n\pi - 1) + \frac{1}{n\pi}(\cos 2n\pi - \cos n\pi) \\ &= \frac{1}{n\pi}(\cos 2n\pi - 2\cos n\pi + 1) \\ &= \frac{1}{n\pi}(1 - 2\cos n\pi + 1) \\ &= \frac{1}{n\pi}(2 - 2\cos n\pi) \\ &= \frac{2}{n\pi}(1 - \cos n\pi) \\ &= \frac{4}{n\pi}\sin^2\frac{n\pi}{2} \\ &= \frac{4}{n\pi}\sin\frac{n\pi}{2} \end{aligned}$$ since $\sin\frac{n\pi}{2}$ alternates between 1, 0, −1

Fourier series: mathematical representation

f(x) written as sum of sinusoidal components :

$$f(x) = \sum_{n=0}^{\infty}\big[\underbrace{A_n\cos(k_n x)}_{\text{even functions}} + \underbrace{B_n\sin(k_n x)}_{\text{odd functions}}\big]$$

$\uparrow$ shape of wave at fixed instant, not time dynamics

Fourier coefficients : $$\boxed{\begin{aligned} A_m &= \frac{2}{\lambda}\int_0^{\lambda} f(x)\cos(k_m x)\,dx \\ B_m &= \frac{2}{\lambda}\int_0^{\lambda} f(x)\sin(k_m x)\,dx \end{aligned} \quad \left(k_n = \frac{2\pi n}{\lambda}\right)}$$

Fourier series : discrete $k_n = \frac{2\pi n}{\lambda}$

Fourier integral : continuous k

as $\lambda \to \infty$, $\Delta k = \frac{2\pi}{\lambda} \to 0$

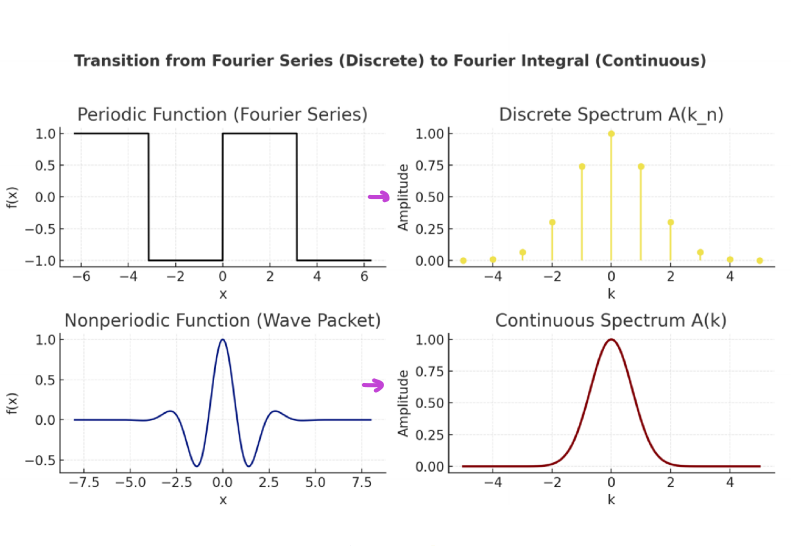
$$\boxed{\begin{aligned} f(x) &= \int A(k)e^{ikx}\,dx \\ A(k) &= \frac{1}{2\pi}\int f(k)\, e^{-ikx}\,dx \end{aligned}}$$ (Fourier transform)

f(x) : position in space (localized in x)

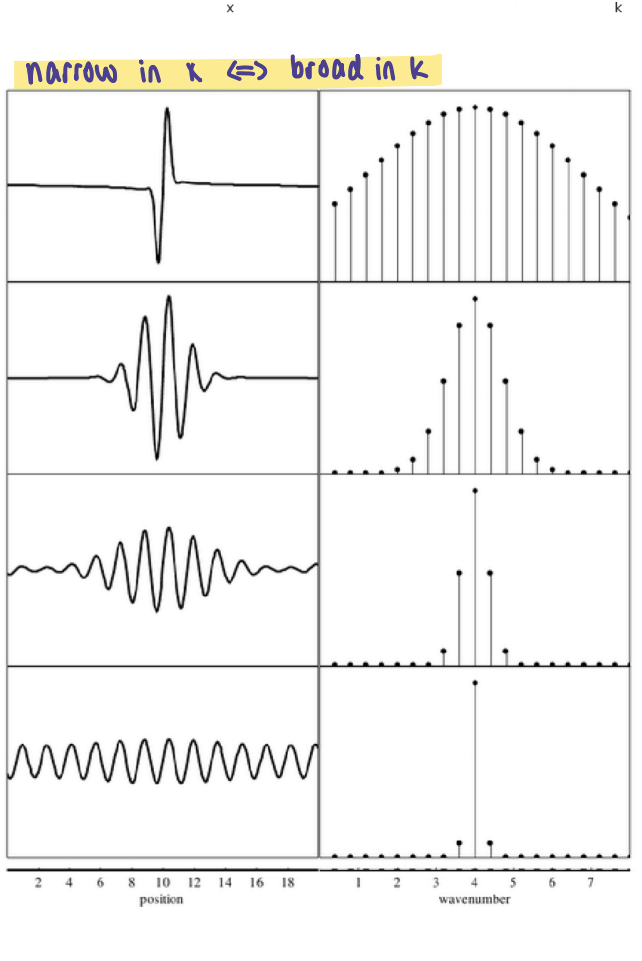
A(k) : wavenumber in space (spread in k)

more localized f(x) = broader A(k) (more Fourier components needed)

Transition from Fourier Series (Discrete) to Fourier Integral (Continuous)



narrow in x $\Leftrightarrow$ broad in k



Fourier uncertainty relation : Δx and Δk denote spreads / uncertainties of wave packet in position and wavenumber space, respectively.

$\Delta x = \sqrt{\langle (x-x_0)^2\rangle}$

$\Delta k = \sqrt{\langle (k-k_0)^2\rangle}$

uncertainty relation : $\boxed{\Delta x \Delta k \geq \frac{1}{2}}$

Time - frequency version of Fourier

angular frequency $\omega = 2\pi f = \frac{2\pi}{T}$

$$\boxed{\begin{aligned} g(t) &= \int_{-\infty}^{\infty} G(\omega)e^{i\omega t}\,d\omega \\ G(\omega) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} g(t)\,e^{-i\omega t}\,dt \end{aligned}}$$

Localization trade-off (uncertainty) : $\boxed{\Delta t \Delta\omega \geq \frac{1}{2}}$

small time width $\Delta t \Leftrightarrow$ large frequency spread $\Delta\omega$

Heisenberg forms : $$\boxed{\begin{aligned} \Delta x\Delta p &\geq \frac{\hbar}{2} \\ \Delta t\Delta E &\geq \frac{\hbar}{2} \end{aligned}}$$

Standing waves → Quantum states

Waveform factorization : $\Psi(x,t) = A\sin(kx)\cos(\omega t) = \underbrace{\Psi(x)}_{\text{spatial}}\underbrace{\cos(\omega t)}_{\text{temporal}}$

$$\underbrace{\Psi(x,t)}_{\text{full}} = \underbrace{\Psi(x)}_{\text{spatial}}\cos(\omega t)$$

General time solution : $f(t) = a\cos(\omega t) + b\sin(\omega t)$

$\Psi(x,t) = \psi(x)[a\cos(\omega t) + b\sin(\omega t)]$ (classical, real)

$\Psi(x,t) = \psi(x)\underbrace{e^{-i\omega t}}_{\cos(\omega t) - i\sin(\omega t)}$ (quantum)

$$\text{Density} = \Psi^*\Psi = |\psi(x)|^2\underbrace{|e^{-i\omega t}|^2}_{} = |\psi(x)|^2$$

$$\begin{aligned} |e^{-i\omega t}|^2 &= |\cos(\omega t) - i\sin(\omega t)|^2 \\ &= \sqrt{\cos^2\omega t + \sin^2\omega t} \\ &= 1 \end{aligned}$$

$\boxed{\text{Density} = |\psi(x)|^2}$ ← independent of time → stationary state

Spatial wave function : $\psi(x) = A\sin(kx)$

Complete wavefunction: $$\begin{aligned} \Psi(x,t) = \psi(x)e^{-i\omega t} &= A\sin(kx)\,e^{-i\omega t} \\ &= A\left(\frac{e^{ikx} - e^{-ikx}}{2i}\right)e^{-i\omega t} \\ &= \frac{A}{2i}\underbrace{\left(e^{i(kx-\omega t)} - e^{-i(kx+\omega t)}\right)}_{\text{superposition of 2 travelling waves}} \end{aligned}$$

$k = \frac{n\pi}{a}$

$\psi(x) = A\sin(kx)$

$\frac{d\psi}{dx} = kA\cos(kx)$

$\frac{d^2\psi}{dx^2} = -k^2A\sin(kx)$

$\boxed{\frac{d^2\psi}{dx^2} = -k^2\psi(x)}$ TISE

$\lambda = \frac{2\pi}{k}$

from de Broglie, $p = \frac{h}{\lambda} = h\left(\frac{k}{2\pi}\right) = 2\pi\hbar\left(\frac{k}{2\pi}\right) = \hbar k$

$\underset{\substack{\uparrow \\ \text{larger momentum = more oscillations /unit}}}{p = \hbar k}$

$K = \frac{p^2}{2m} = \frac{\hbar^2k^2}{2m}$

$\therefore \frac{d^2\psi}{dx^2} = -\frac{2mK}{\hbar^2}\psi$

$\underset{\substack{\uparrow \\ \text{larger changes = higher KE}}}{\frac{d^2\psi}{dx^2}} = -\frac{2m}{\hbar^2}(E - U(x))\psi$, since $E = K + U$

$$\boxed{-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + U(x)\psi = E\psi}$$ ← Time-independent Schrödinger equation

↳ $\hat{H}\psi = E\psi$

Ex.

$\psi(x) = e^{ikx}$	$\psi_2(x) = \cos(kx)$	$\psi_3(x) = \sin(kx)$	$\psi_4(x) = Ae^{ikx} + Be^{-ikx}$
$\psi'(x) = ike^{ikx}$	$\psi_2'(x) = -k\sin(kx)$	$\psi_3'(x) = k\cos(kx)$	$\psi_4'(x) = ikAe^{ikx} - ikBe^{-ikx}$
$\psi''(x) = -k^2e^{ikx}$	$\psi_2''(x) = -k^2\cos(kx)$	$\psi_3''(x) = -k^2\sin(kx)$	$\psi_4''(x) = -k^2Ae^{ikx} - k^2Be^{-ikx}$
$= -k^2\psi(x)$	$= -k^2\psi(x)$	$= -k^2\psi(x)$	$= -k^2\psi(x)$
satisfies TISE	satisfies TISE	satisfies TISE	satisfies TISE

Every measurable quantity is represented by operator acting on ψ

Hamiltonian operator : $\hat{H} = \underbrace{-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}}_{\text{kinetic}} + \underbrace{U(x)}_{\text{potential}}$

energy : $\underset{\substack{\uparrow \\ \text{operator}}}{\hat{H}}\psi(x) = \underset{\substack{\uparrow \\ \text{energy eigenvalue}}}{E}\psi(x)$

position :	$\hat{x} = x$	multiplication, spatial location
momentum :	$\hat{p} = -i\hbar\frac{d}{dx}$	change in spatial location
kinetic energy :	$\hat{T} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2}$	derived from momentum
potential energy :	$\hat{U} = U(x)$	multiplication, depends on position
total energy :	$\hat{H} = \hat{T} + \hat{U} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + U(x)$	Hamiltonian, Schrödinger's equation

probability distribution : $P(x) = |\psi(x)|^2$

expectation value of position : $\langle x\rangle = \int x|\psi(x)|^2\,dx$

expectation value : $\boxed{\langle\hat{A}\rangle = \int_{-\infty}^{\infty} \underbrace{\psi^*(x)}_{\text{conjugate}}\,\hat{A}\psi(x)\,dx}$

Ex. $\psi(x) = Ae^{-x/L}$, $x \geq 0$

$\int_{-\infty}^{\infty} |\psi(x)|^2\,dx = 1$

$\int_0^{\infty} A^2e^{-2x/L}\,dx = 1$

$A^2\left[-\frac{L}{2}e^{-2x/L}\right]_0^{\infty} = 1$

$A^2\left(\frac{L}{2}\right) = 1$

$A = \sqrt{\frac{2}{L}}$

prob. density : $|\psi(x)|^2 = \frac{2}{L}e^{-2x/L}$, $x \geq 0$

expectation value of momentum: $$\begin{aligned} \langle p\rangle &= \int_0^{\infty} Ae^{-x/L}(-i\hbar)\frac{d}{dx}(Ae^{-x/L})\,dx \\ &= \int_0^{\infty} Ae^{-x/L}(-i\hbar)\left(-\frac{1}{L}Ae^{-x/L}\right)dx \\ &= \int_0^{\infty} \frac{i\hbar}{L}A^2e^{-2x/L}\,dx \\ &= \frac{i\hbar}{L}A^2\left[-\frac{L}{2}e^{-2x/L}\right]_0^{\infty} \\ &= \frac{i\hbar A^2}{2} \\ &= \frac{i\hbar}{2}\left(\frac{2}{L}\right) \\ &= \frac{i\hbar}{L} \end{aligned}$$ not physical, since $\langle p\rangle$ is imaginary

$\frac{d}{dx}Ae^{-x/L} = -\frac{1}{L}Ae^{-x/L}$

$\frac{d^2}{dx^2}Ae^{-x/L} = \frac{1}{L^2}Ae^{-x/L}$

expectation value of energy: $\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2}$

$$\begin{aligned} \langle E\rangle &= \int_0^{\infty}(Ae^{-x/L})\left(-\frac{\hbar^2}{2m}\right)\left(\frac{d^2}{dx^2}Ae^{-x/L}\right)dx \\ &= \left(-\frac{1}{L^2}A^2\right)\left(-\frac{\hbar^2}{2m}\right)\int_0^{\infty}e^{-2x/L}\,dx \\ &= \left(-\frac{1}{L^2}A^2\right)\left(-\frac{\hbar^2}{2m}\right)\left[-\frac{L}{2}e^{-2x/L}\right]_0^{\infty} \\ &= \left(\frac{1}{2L}A^2\right)\left(-\frac{\hbar^2}{2m}\right) \\ &= \left(\frac{1}{2L}\cdot\frac{2}{L}\right)\left(-\frac{\hbar^2}{2m}\right) \\ &= -\frac{\hbar^2}{2mL^2} \end{aligned}$$ unphysical, since KE cannot be −ve

Time dependent Schrödinger eqn

$\Psi(x,t) = \psi(x)e^{-i\omega t} = \psi(x)e^{-iEt/\hbar}$

$\frac{\partial}{\partial t}$: $\frac{\partial\Psi}{\partial t} = -\frac{iE}{\hbar}\psi(x)e^{-iEt/\hbar} = -\frac{iE}{\hbar}\Psi(x,t)$

$\times i\hbar$: $i\hbar\frac{\partial\Psi}{\partial t} = E\Psi(x,t)$

Time - independent Schrödinger eqn:

$E\psi(x) = \left[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + U(x)\right]\psi(x)$

multiply both sides by $e^{-iEt/\hbar}$:

$\underbrace{E\psi(x)e^{-iEt/\hbar}}_{\Psi(x,t)} = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + U(x)\right]\underbrace{\psi(x)e^{-iEt/\hbar}}_{\Psi(x,t)}$

$$\boxed{i\hbar\frac{\partial\Psi(x,t)}{\partial t} = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + U(x)\right]\Psi(x,t)}$$ ← Time-dependent Schrödinger eqn.

↳ $i\hbar\frac{\partial\Psi(x,t)}{\partial t} = \hat{H}\Psi(x,t)$

Time - independent : $\hat{H}\psi(x) = E\psi(x)$ possible energy states

Time - dependent : $\hat{H}\Psi(x,t) = i\hbar\frac{\partial\Psi(x,t)}{\partial t}$ how states evolve over time

For a stationary eigenstate : $\Psi(x,t) = \psi(x)\underline{e^{-iEt/\hbar}}$

ensures each stationary state evolves with fixed phase freq. proportional to energy ($\omega = E/\hbar$)

Ex. Calculate prob. density of stationary state $\Psi(x,t) = \psi(x)e^{-iEt/\hbar}$

$$\begin{aligned} |\Psi_n(x,t)|^2 &= \Psi_n^*(x,t)\Psi_n(x,t) \\ &= \left[\psi_n^*(x)e^{iEt/\hbar}\right]\left[\psi_n(x)e^{-iEt/\hbar}\right] \\ &= \psi_n^*(x)\psi_n(x)e^0 \\ &= |\psi_n(x)|^2 \end{aligned}$$

conjugate of e^{ix} is just e^{-ix}

Ex. Calculate prob. density of superposition wavefunction

$\Psi(x,t) = c_1\psi_1(x,t)e^{-iE_1t/\hbar} + c_2\psi_2(x)e^{-iE_2t/\hbar}$

$$\begin{aligned} |\Psi(x,t)|^2 &= |c_1|^2|\psi_1(x,t)|^2 + |c_2|^2|\psi_2(x,t)|^2 + \underline{c_1^*\psi_1^*(x)e^{E_1t/\hbar}\,c_2\psi_2(x)e^{-E_2t/\hbar}} + \underline{c_2^*\psi_2^*(x)e^{E_2t/\hbar}\,c_1\psi_1(x)e^{-E_1t/\hbar}} \\ &= |c_1|^2|\psi_1(x,t)|^2 + |c_2|^2|\psi_2(x,t)|^2 + 2\,\mathrm{Re}\left[c_1c_2^*\psi_1(x)\psi_2^*(x)e^{-i(E_1-E_2)t/\hbar}\right] \end{aligned}$$

complex conjugates ⇒ sum = 2× real part

Superposition

- particle can be in multiple states at once until measured
- superposition = comb. of probabilities

eigenfunctions from Schrödinger's eqn $\{\psi_n\}$ w/ energy E_n

well- behaved system satisfies:

① **orthogonal :** $\int\psi_m^*(x)\psi_n(x)\,dx = 0$ independence

② **normalized :** $\int|\psi_n(x)|^2\,dx = 1$ physical meaning

⇒ **orthonormal condition :** $\boxed{\int\psi_m^*(x)\psi_n(x)\,dx = \delta_{mn}}$

completeness : set $\{\psi_n\}$ is complete if any physically valid wavefunction $\psi(x)$ can be expressed as linear comb. of them

$\psi(x) = \sum_n c_n\psi_n(x)$

means function spans Hilbert space

$\psi(x) = \sum_n c_n\psi_n(x)$

Isolate each coefficient : multiply ψ^*_m

$\int\psi^*_m(x)\psi(x)\,dx = \sum_n c_n\underline{\int\psi_m^*(x)\psi_n(x)\,dx}$

0 if $m \neq n$, 1 if $m = n$ by orthonormality

$$\boxed{c_n = \int\psi_n^*(x)\psi(x)\,dx}$$

$|c_n|^2$: probability of energy E_n

Normalization: $$\begin{aligned} \int|\psi(x)|^2\,dx &= \int\psi^*(x)\psi(x)\,dx \\ &= \sum_{m,n}c_m^*c_n\int\psi_m^*(x)\psi_n(x)\,dx \\ &= |c_n|^2 = 1 \end{aligned}$$

Probability density : $\boxed{\underset{\substack{P(x),\ \text{position prob.}}}{|\psi(x)|^2} = \psi^*(x)\psi(x) = \sum_{m,n}c_m^*c_n\psi_m^*(x)\psi_n(x)}$

Expectation value: $\boxed{\langle E\rangle = \sum_n|c_n|^2E_n}$

$\Psi(x,t) = \psi(x)e^{-iEt/\hbar}$ (time -dependent)

$\Psi(x,t) = \sum_n c_n\underbrace{\psi_n(x)}_{\text{spatial}}\underbrace{e^{-iE_nt/\hbar}}_{\text{temporal}}$ (phase evolution)

spatial (where it is likely to be found)

$$\begin{aligned} \int|\Psi(x,t)|^2\,dx &= \int\Psi^*(x,t)\Psi(x,t)\,dx = \int\sum_{n,m}c_n^*c_m\psi_n^*(x)\psi_m(x)e^{i(E_n-E_m)t/\hbar}\,dx \\ &= \int|\Psi(x,t)|^2\,dx = \sum_n|c_n|^2 = 1 \end{aligned}$$

Ex. $\psi_n(x) = \sqrt{\frac{2}{L}}\sin\frac{n\pi x}{L}$, $E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}$, $0 < x < L$

Consider superposition $\Psi(x,0) = \underset{c_1}{\frac{1}{\sqrt{3}}}\underset{\psi_1}{\psi_1(x)} + \underset{c_2}{\sqrt{\frac{2}{3}}e^{i\phi}}\underset{\psi_2}{\psi_2(x)}$

$\int_0^L|\Psi(x,0)|^2 = \left|\frac{1}{\sqrt{3}}\right|^2 + \left|\frac{\sqrt{2}}{3}e^{i\phi}\right|^2 = \frac{1}{3} + \frac{2}{3} = 1$

$$\begin{aligned} \Psi(x,t) &= \frac{1}{\sqrt{3}}\psi_1(x)e^{-iE_1t/\hbar} + \sqrt{\frac{2}{3}}e^{i\phi}\psi_2(x)e^{-iE_2t/\hbar} \\ &= \frac{1}{\sqrt{3}}\sqrt{\frac{2}{L}}\sin\frac{\pi x}{L}e^{-iE_1t/\hbar} + \sqrt{\frac{2}{3}}\sqrt{\frac{2}{L}}e^{i\phi}\sin\frac{2\pi x}{L}e^{-iE_2t/\hbar} \end{aligned}$$

$P(E_1) = \left|\frac{1}{\sqrt{3}}\right|^2 = \frac{1}{3}$

$P(E_2) = \left|\sqrt{\frac{2}{3}}\right|^2 = \frac{2}{3}$

$$\begin{aligned} \langle E\rangle &= \sum_n|c_n|^2E_n \\ &= \frac{1}{3}E_1 + \frac{2}{3}E_2 \\ &= \frac{1}{3}\frac{\pi^2\hbar^2}{2mL^2} + \frac{2}{3}\frac{4\pi^2\hbar^2}{2mL^2} \end{aligned}$$

Free particle (V(x) = 0 everywhere)

$-\frac{2mE}{\hbar^2} = -k^2$, $k = \sqrt{2mE/\hbar^2}$

Schrodinger : $-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi$

$\frac{d^2\psi}{dx^2} = -k^2\psi$

general soln : $\psi(x) = A\sin(kx) + B\cos(kx)$, $k = \sqrt{\frac{2mE}{\hbar^2}}$

$= \underset{\substack{\text{"right"} \\ p = \hbar k}}{Ce^{ikx}} + \underset{\substack{\text{"left"} \\ p = -\hbar k}}{De^{-ikx}}$ (Time- independent)

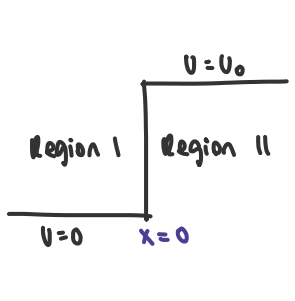
$\Psi(x,t) = (Ce^{ikx} + De^{-ikx})e^{-\frac{iEt}{\hbar}}$ (Time - dependent)

$= Ce^{i(kx-\omega t)} + De^{-i(kx+\omega t)}$, $\omega = \frac{E}{\hbar}$

opposite momentum components

Potential step

$$U(x) = \begin{cases} 0, & x < 0 \\ U_0, & x > 0 \end{cases}$$



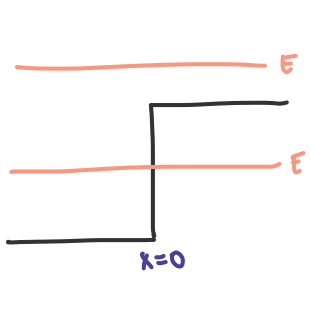
① $E > U_0$: particle has enough energy, partial reflection / transmission, oscillatory

② $E < U_0$: classically forbidden region, tunnelling decay

both $E > U_0$ and $E < U_0$

Region I (x < 0) : $\psi_1(x) = \underset{\substack{\text{incident} \\ \text{"right"}}}{Ae^{ik_1x}} + \underset{\substack{\text{reflected} \\ \text{"left"}}}{Be^{-ik_1x}}$, $k_1 = \frac{\sqrt{2mE}}{\hbar}$

$E > 0$

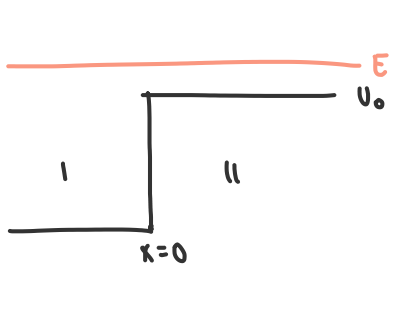


boundary condition : ψ_1 and ψ_1' are continuous at $x = 0$

Region II (x > 0) : $E > U_0$: $\psi_2(x) = Ce^{ik_2x} + \cancel{De^{-ik_2x}}$, $k_2 = \sqrt{\frac{2m(E-U_0)}{\hbar}}$

$E - U_0 > 0$

only keep right- moving term, since $x \to +\infty$ (no left-moving wave in region II)



$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = (E-U_0)\psi \Rightarrow \frac{d^2\psi}{dx^2} = -\frac{2m(E-U_0)}{\hbar^2}\psi$

$k_2 = \sqrt{\frac{2m(E-U_0)}{\hbar^2}}$

boundary conditions: $\psi_1(0) = \psi_2(0)$

$Ae^{ik_1x} + Be^{-ik_1x} = Ce^{ik_2x} \Rightarrow \underline{A + B = C}$

$ik_1Ae^{ik_1x} - ik_1Be^{-ik_1x} = ik_2Ce^{ik_2x} \Rightarrow \underline{k_1(A-B) = k_2C}$

⇒ get $\frac{B}{A}$, $\frac{C}{A}$

$$\boxed{\begin{aligned} R &= \frac{\text{Reflected}}{\text{Incident}} = \left|\frac{B}{A}\right|^2 = \left(\frac{k_1-k_2}{k_1+k_2}\right)^2 \\ T &= \frac{\text{Transmitted}}{\text{Incident}} = \left|\frac{C}{A}\right|^2 = \frac{4k_1k_2}{(k_1+k_2)^2} \end{aligned}}$$

$A + B = \frac{k_1}{k_2}(A - B)$

$\frac{k_2-k_1}{k_2}A = -\frac{k_2-k_1}{k_2}B$

$(k_1-k_2)A = (k_1+k_2)B$

$k_1(A - C + A) = k_2C$

$2Ak_1 - Ck_1 = k_2C$

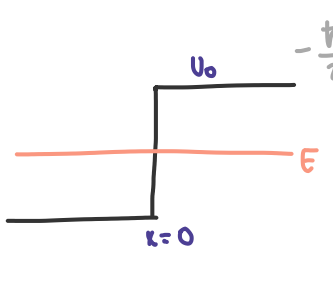
$2Ak_1 = (k_1+k_2)C$

$\frac{C}{A} = \frac{2k_1}{k_1+k_2}$

$E < U_0$: $\psi_2(x) = Ce^{-\alpha x} + \cancel{De^{\alpha x}}$, $\alpha = \sqrt{\frac{2m(U_0-E)}{\hbar}}$

$U_0 - E > 0$

diverges as $x \to +\infty$



$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = (U_0-E)\psi$

$\frac{d^2\psi}{dx^2} = -\frac{2m(U_0-E)}{\hbar^2}\psi$

$\alpha = \frac{\sqrt{2m(U_0-E)}}{\hbar}$

boundary conditions: $\psi_1(0) = \psi_2(0)$

$Ae^{ik_1x} + Be^{-ik_1x} = Ce^{-\alpha x} \Rightarrow A + B = C$

$ik_1Ae^{ik_1x} - ik_1Be^{-ik_1x} = -\alpha Ce^{-\alpha x} \Rightarrow ik_1(A-B) = -\alpha C \Rightarrow k_1(A-B) = i\alpha C$

$A + B = \frac{k_1(A-B)}{i\alpha}$

$i\alpha A + i\alpha B = k_1A - k_1B$

$(i\alpha - k_1)A = (-i\alpha - k_1)B$

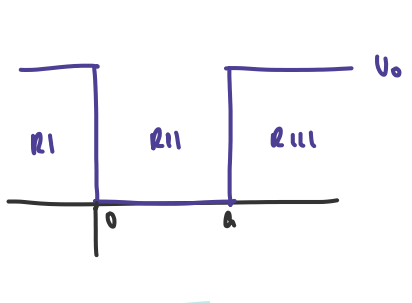
$\frac{B}{A} = \frac{k_1 - i\alpha}{k_1 + i\alpha}$

$R = \left|\frac{B}{A}\right|^2 = \left|\frac{k_1 - i\alpha}{k_1 + i\alpha}\right|^2 = 1$

no prob passes potential step to $+\infty$ but particle can enter region II (decays into region II)

Finite well

finite potential barriers – wavefunction leaks out



$$U(x) = \begin{cases} U_0, & x < 0 \\ 0, & 0 < x < a \\ U_0, & x > a \end{cases}$$

Region II : $U(x) = 0$, $E > 0$ oscillatory, like infinite well

$\psi_2(x) = A\sin(kx) + B\cos(kx)$

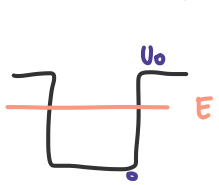
$= Ce^{ikx} + De^{-ikx}$, $k = \frac{\sqrt{2mE}}{\hbar}$

$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi$

$\frac{d^2\psi}{dx^2} = -\frac{2mE}{\hbar^2}\psi$

$k = \frac{\sqrt{2mE}}{\hbar}$

Region I : left of well $(x < 0)$



$E < U_0$: $U_0 - E > 0$

$\psi_1(x) = Ae^{\alpha x} + \cancel{Be^{-\alpha x}}$, $\alpha = \frac{\sqrt{2m(U_0-E)}}{\hbar}$

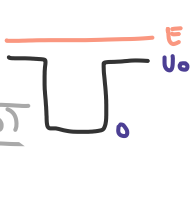
must be finite as $x \to -\infty$

$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = (U_0-E)\psi$

$\alpha = \frac{\sqrt{2m(U_0-E)}}{\hbar}$

$E > U_0$: $E - U_0 > 0$

$\psi_1(x) = Ce^{ikx} + De^{-ikx}$, $k = \frac{\sqrt{2m(E-U_0)}}{\hbar}$



Region 3 : Right of well $(x > a)$

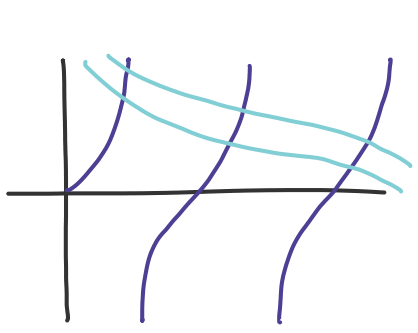
$E < U_0$: $U_0 - E > 0$

$\psi_3(x) = \cancel{Ae^{\alpha x}} + Be^{-\alpha x}$, $\alpha = \frac{\sqrt{2m(U_0-E)}}{\hbar}$

must be finite as $x \to +\infty$

$\psi_3(x) = Ge^{-\alpha x}$, $\alpha = \frac{\sqrt{2m(U_0-E)}}{\hbar}$

After applying boundary conditions, $\tan(ka) = \frac{2\alpha k}{k^2 - \alpha^2}$, where $k = \frac{\sqrt{2mE}}{\hbar}$, $\alpha = \frac{\sqrt{2m(U_0-E)}}{\hbar}$



finite number of solns ⇒ finite number of bound energy levels

k_n found : $E_n = \frac{\hbar^2k_n^2}{2m}$